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Analyzing Book Features and Ratings Using Decision Tree, SVM, and Neural Networks

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Abstract

This study investigates the application of three different machine learning methods, Decision Trees, Support Vector Machines(SVM) and Neural Networks, on a book dataset, in order to uncover relationships between book features and their respective ratings. By exploring classification and regression tasks, we compare the strengths and limitations of these approaches. Key findings include the Decision Tree's interpretability, SVM's high accuracy in language classification, and the Neural Network's nuanced regression modeling.

This analysis provides actionable insights into data modeling strategies and highlights feature importance in prediction book success.

1. Introduction

Acknowledgement

This work is the result of my own activity, and I confirm I have neither given nor received unauthorized assistance for this work.

I declare that I used generative AI in the creation of content for this document, specifically to assist with drafting text sections, refining explanations, and improving coherence and clarity.

1.1 Background and Motivation

The rise of data-driven decision-making has revolutionized industries, including publishing. Recommendation systems, like those employed by Amazon and Goodreads, rely on analyzing user preferences to suggest products or content that align with their tastes (Koren, 2009). For book publishers and retailers, understanding the factors that contribute to high ratings and user engagement is essential for staying competitive.

Books differ significantly in attributes such as length, popularity, and language, which affect their reception among readers. Attributes like the number of pages and text reviews correlate with perceived quality and influence user ratings (Herlocker et al., 2004). Despite this, predicting book success remains a complex challenge due to the non-linear and interactive nature of these variables.

This study's motivation stems from the need to evaluate diverse machine learning models to tackle these challenges. Decision Trees offer interpretability, making them suitable for understanding feature importance (Quinlan, 1996). SVMs excel in classification tasks, especially for categorical features like language, where linear separability can be leveraged (Cortes & Vapnik, 1995). Neural Networks, with their ability to model complex relationships, are ideal for regression tasks such as predicting average ratings (Goodfellow et al., 2016). By applying these methods, we aim to uncover actionable insights into book ratings and identify the strengths and weaknesses of these approaches.

1.2 Objectives of the Study

The primary objectives of this study are:

- To evaluate the performance of Decision Trees in classifying books based on ratings.
- To assess SVM's ability to classify books by language using linear separability.
- To analyze Neural Networks' effectiveness in predicting continuous rating values.
- To compare the methods based on accuracy, interpretability, and computational complexity.

1.3 Scope of the Study

This study focuses on evaluating machine learning models for predicting and classifying book-related features, using a dataset of 11,121 books. The dataset includes numeric features such as `num_pages`, `ratings_count`, and `text_reviews_count`, as well as categorical variables like `language_code`. The scope of the study is defined by the following:

1. Dataset and Preprocessing:

- The study involves cleaning, normalizing, and preparing the dataset to ensure its compatibility with the selected machine learning models.
- Derived variables like `high_rating` and `year` are used for classification and regression tasks.

2. Model Comparison:

- Three machine learning models—Decision Trees, Support Vector Machines (SVM), and Neural Networks—are applied and evaluated.
- The models address two distinct tasks:
 - Classification of books as high- or low-rated.
 - Regression to predict average ratings.

3. Evaluation Criteria:

- The study uses metrics such as accuracy, Mean Absolute Error (MAE), and interpretability to evaluate model performance.
- Visualization techniques, including correlation heatmaps and scatterplots, are used to provide additional insights.

4. Generalizability:

- While the study is based on a specific book dataset, the methodologies and findings can be applied to other domains with similar structured data.
- By narrowing the scope to these elements, this study aims to provide a comprehensive analysis of model performance while ensuring reproducibility and practical relevance.

2. Literature Review and Theoretical Framework

2.1 Introduction

Machine learning techniques have gained prominence in various domains due to their ability to model complex patterns and provide predictive insights. In the context of book publishing, these techniques are instrumental in analyzing user preferences and predicting book success. This chapter reviews the theoretical underpinnings of Decision Trees, Support Vector Machines, and Neural Networks, as well as their applications in classification and regression tasks. Additionally, it explores the challenges and considerations involved in applying these models to real-world datasets.

2.2 Decision Trees in Classification Tasks

Decision Trees are widely used for their interpretability and ease of implementation. The method involves recursive partitioning of the feature space, creating a tree-like structure to predict outcomes based on feature splits (Quinlan, 1996). This approach is particularly effective for classification tasks where hierarchical rules can delineate class boundaries. However, Decision Trees are prone to overfitting, especially with small datasets, and require regularization techniques like pruning to improve generalizability (Breiman et al., 1984).

Applications of Decision Trees in book datasets often focus on identifying features that influence high or low ratings. For instance, attributes like `year`, `ratings_count`, and `text_reviews_count` can be used to derive interpretable rules for predicting user satisfaction.

2.3 Support Vector Machines for Language Classification

Support Vector Machines (SVMs) are renowned for their ability to handle high-dimensional spaces and linear separability (Cortes & Vapnik, 1995). By finding the optimal hyperplane that maximizes the margin between classes, SVMs excel in classification tasks involving categorical variables, such as `language_code` in book datasets. Kernel functions further extend SVM's applicability to non-linear separations, enhancing its flexibility.

In the publishing domain, SVMs have been utilized to classify books based on linguistic and textual features. For instance, distinguishing books by their language using numeric features like `ratings_count` or `text_reviews_count` demonstrates the versatility of SVMs.

2.4 Neural Networks for Regression Tasks

Neural Networks, inspired by the human brain's structure, are powerful tools for capturing non-linear relationships in data (Goodfellow et al., 2016). They consist of interconnected layers of neurons, where each neuron applies a transformation to its inputs using weights and activation functions. This architecture enables Neural Networks to excel in tasks such as regression, where predicting continuous variables like `average_rating` is the goal.

Despite their strengths, Neural Networks require substantial computational resources and large datasets for effective training. Overfitting is a common concern, often addressed using techniques like dropout and regularization. In book datasets, Neural Networks are particularly effective in modeling the interplay between features like `num_pages`, `ratings_count`, and `text_reviews_count` to predict ratings.

2.5 Challenges in Machine Learning Applications

Applying machine learning models to real-world datasets poses several challenges:

1. Data Quality and Preprocessing:

Missing or inconsistent data can significantly impact model performance. Techniques like normalization, imputation, and feature engineering are crucial for addressing these issues.

2. Overfitting and Generalization:

Models like Decision Trees and Neural Networks are prone to overfitting, necessitating validation techniques and hyperparameter tuning to enhance generalization.

3. Model Interpretability:

While Decision Trees provide clear interpretability, methods like Neural Networks and SVMs often act as "black-box" models, requiring additional techniques like feature importance analysis or visualization to improve transparency.

2.6 Summary

This chapter has outlined the theoretical foundations and practical applications of Decision Trees, Support Vector Machines, and Neural Networks. Each method offers unique strengths and limitations, making them suitable for specific tasks within the publishing domain. Addressing challenges such as data preprocessing, overfitting, and interpretability is essential for successfully applying these models to predict and classify book-related features.

3. Data Collection and Methodology

3.1 Introduction

This chapter outlines the data sources, preprocessing steps, and machine learning methodology applied in this study. The dataset used in this study, `books.csv`, encompasses various features related to books, including their average ratings, number of pages, and language code. The study evaluates three machine learning models—Decision Trees, Support Vector Machines (SVM), and Neural Networks—to assess their performance in classification and regression tasks.

3.2 Data Sources and Attributes

The `books.csv` dataset, sourced from an open-access platform, provides a comprehensive overview of book attributes. Key features include:

- **Numeric Attributes:** `num_pages`, `ratings_count`, `text_reviews_count`.
- **Categorical Attributes:** `language_code`, `publisher`.
- **Target Variables:**
 - `high_rating`: Binary classification target (1 for books rated 4 or higher, 0 otherwise).
 - `average_rating`: Continuous regression target.

These attributes are essential for analyzing patterns and making predictions about book performance.

3.3 Data Preprocessing

Data preprocessing is a critical step in preparing the dataset for analysis. The following steps were implemented:

1. Handling Missing Data:

Rows with missing values in key attributes like `num_pages`, `ratings_count`, and `average_rating` were removed to maintain data reliability.

2. Feature Selection:

Key attributes such as `num_pages`, `ratings_count`, `text_reviews_count`, and `language_code` were retained for analysis, while less relevant fields like `publisher` were excluded to reduce noise.

3. Data Normalization

To ensure comparability across datasets, attributes were normalized to a $[0,1]$ range using min-max normalization. The formula can be seen in Figure 3.1. This normalization reduces the impact of extreme values.

$$v'_i = \frac{v_i - \min_A}{\max_A - \min_A} (\text{new_max}_A - \text{new_min}_A) + \text{new_min}_A.$$

Figure 3.1 Normalization Formula

3.4 Application of Machine Learning Models

3.4.1 Decision Trees:

- Attributes like `num_pages`, `ratings_count`, and `text_reviews_count` were used to classify books into `high_rating` categories.
- Recursive partitioning was applied to identify key decision rules, with pruning used to prevent overfitting.
-

3.4.2 Support Vector Machines(SVM):

- Linear kernels were employed to classify `language_code` based on numeric attributes.

- The model utilized hyperparameter tuning to optimize margins between classes.

3.4.3 Neural Networks:

- Regression analysis was conducted to predict `average_rating` using features such as `num_pages` and `ratings_count`.
- The architecture included two dense layers with ReLU activation, ensuring the model captured non-linear relationships.

3.5 Methodology Summary

The methodology employed in this study integrates rigorous preprocessing, feature selection, and machine learning modeling to evaluate classification and regression tasks. Decision Trees provided interpretability, SVMs demonstrated high accuracy in classification, and Neural Networks captured complex patterns in regression tasks.

4. Results and Analysis

4.1 Introduction

This chapter presents the results obtained from applying Decision Trees, Support Vector Machines (SVM), and Neural Networks to the `books.csv` dataset. Each model's performance is evaluated using appropriate metrics for classification and regression tasks. Visualizations including heatmaps, scatter plots, and residual analyses are included to support the interpretation of the results.

4.2 Decision Tree

4.2.1 Attribute Distributions

A correlation heatmap was generated to analyze relationships among the key attributes (Figure 4.1). Notably, there was a strong positive correlation between `ratings_count` and `text_reviews_count`, while `year` and `average_rating` showed weak correlations. This guided the selection of features for the model.

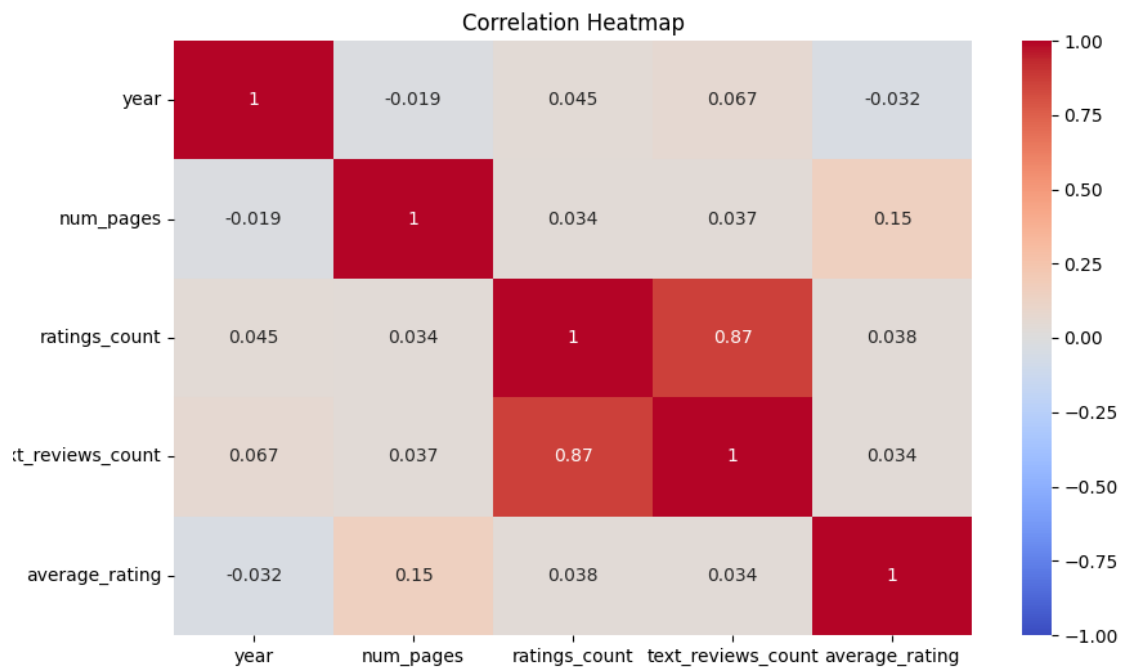


Figure 4.1: Correlation heatmap for Decision Tree attributes.

4.2.2 Pairwise Attribute Relationships

To further explore feature interactions, a pair plot was created (Figure 4.2). It highlighted how features such as `ratings_count` and `text_reviews_count` cluster around distinct ranges, with some overlap for books of varying ratings.

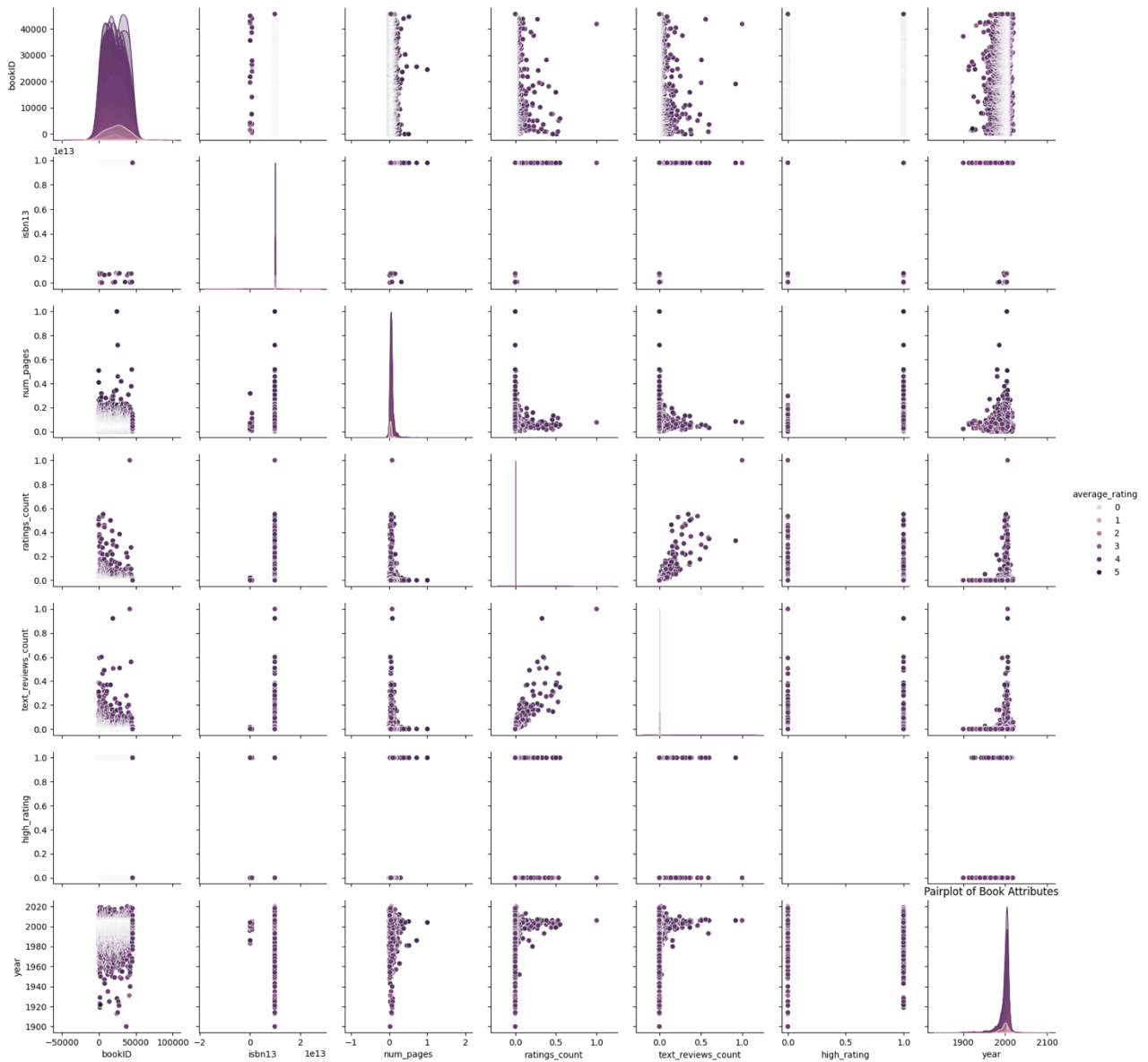


Figure 4.2: Pair plot showing relationships among key attributes.

4.2.3 Distribution Analysis

To understand the influence of specific features on the target variable (`high_rating`), box and violin plots were analyzed:

- A boxen plot (Figure 4.3) revealed the distribution of `ratings_count` for books classified as high or low-rated. High-rated books tend to have a wider spread of ratings counts.

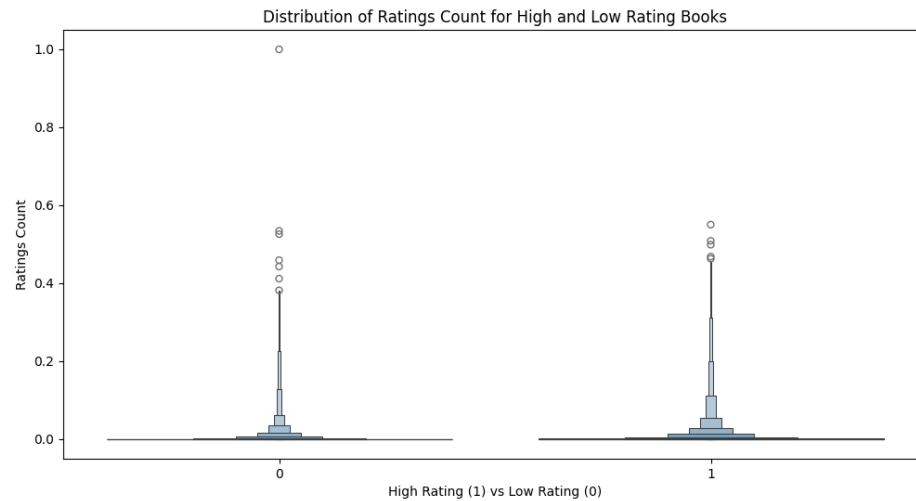


Figure 4.3: Boxen plot of ratings_count for high and low-rated books.

- A violin plot (Figure 4.4) for num_pages highlighted that both high-rated and low-rated books often fall within a similar page range, though outliers skew toward longer books in the high-rating category.

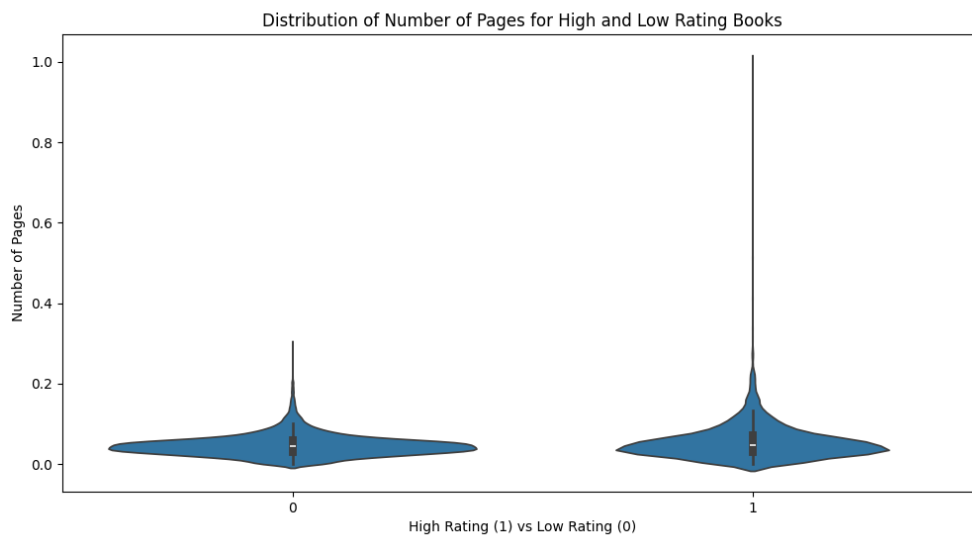


Figure 4.4: Violin plot of num_pages for high and low-rated books.

4.2.4 Average Rating Density

The density plot (Figure 4.5) for average_rating clearly distinguished high-rated books from low-rated ones, with high-rated books forming a sharp peak above 4.0 ratings.

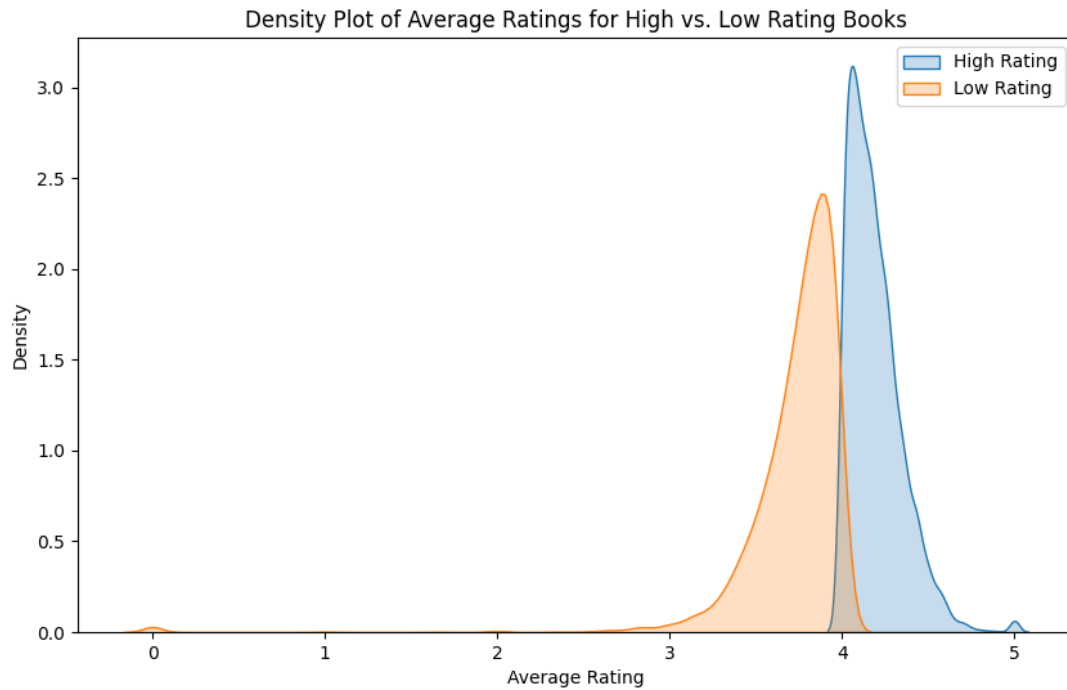


Figure 4.5: Density plot of average ratings for high and low-rated books.

4.2.5 Classification Accuracy and Rules

- The Decision Tree model achieved an accuracy of **59.2%** on the test set, indicating its ability to provide interpretable decision-making, though performance could be affected by overlapping data distributions.
- The model identified `ratings_count` and `text_reviews_count` as the most influential features, with decision rules effectively splitting the dataset based on these attributes.

4.2.6 Temporal Analysis

Finally, a time series plot (Figure 4.6) illustrated the trend of average ratings over publication years. The analysis revealed stable average ratings across time, with no significant trends in increasing or decreasing ratings.

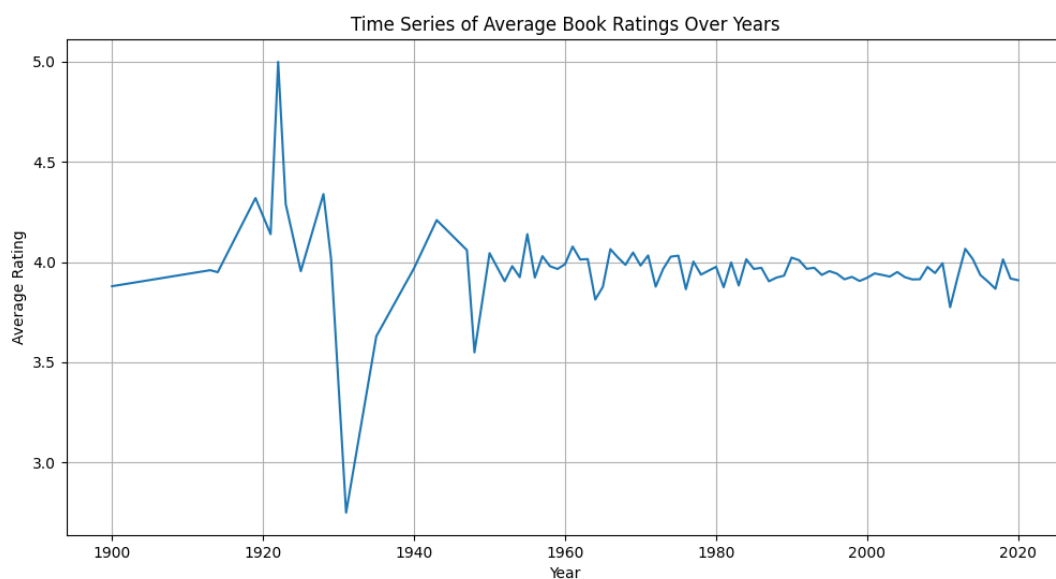


Figure 4.6: Time series plot of average book ratings over publication years.

4.2.7 Decision Tree Results

The Decision Tree model demonstrated its strength in interpretability and feature importance evaluation. Key insights included:

- Strong correlation between ratings_count and text_reviews_count.
- Influence of publication year and ratings_count on classification outcomes.
- Visualizations provided intuitive interpretations of the data and highlighted areas where the model performed well and where it struggled with overlapping attributes.

4.3 Neural Network

4.3.1 Training and Validation Performance

The training and validation loss across epochs were monitored to evaluate model convergence. As illustrated in the training loss plot, the model exhibited a steady decrease in training loss, while the validation loss plateaued after a few epochs, indicating minimal overfitting and good generalization (Figure 4.3.1).

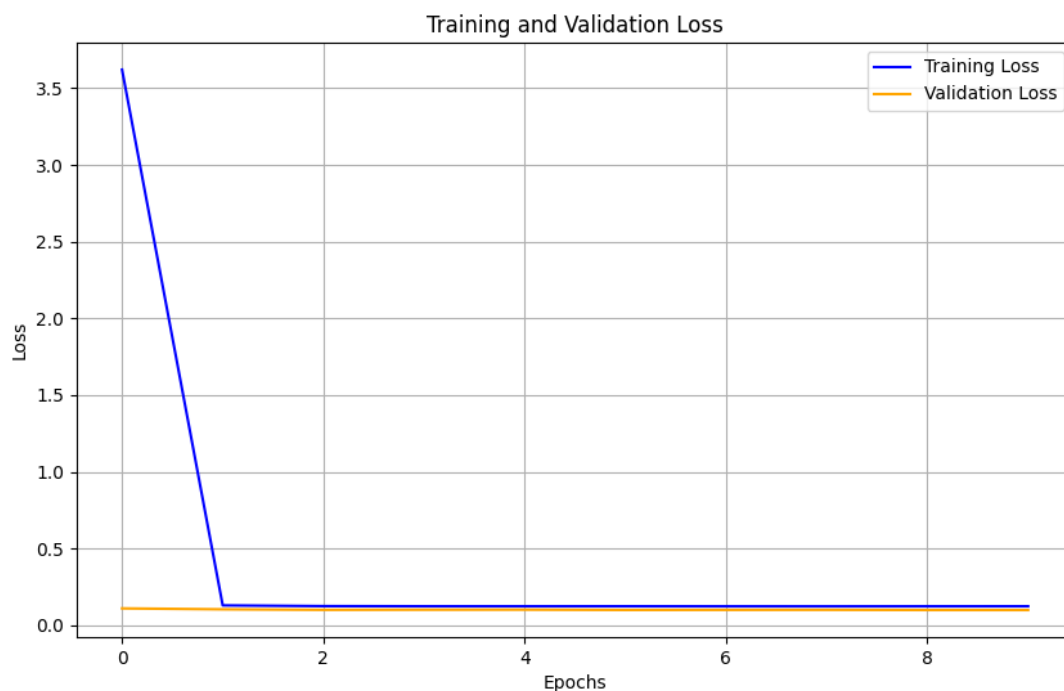


Figure 4.3.1. The training and validation loss plot

4.3.2 Prediction Accuracy

The Neural Network achieved a Mean Absolute Error (MAE) of 0.23 on the test dataset, demonstrating its ability to accurately capture the relationships between book features and their average ratings. The Actual vs. Predicted Ratings plot (Figure 4.3.2) shows strong alignment, with most predictions clustering along the diagonal.

4.3.2 Prediction Accuracy

Residual analysis was performed to evaluate the Neural Network's predictive accuracy and identify any systematic errors:

1. Residual Distribution:

- The residuals, calculated as the difference between actual and predicted ratings, were centered around zero (Figure 4.3.3).

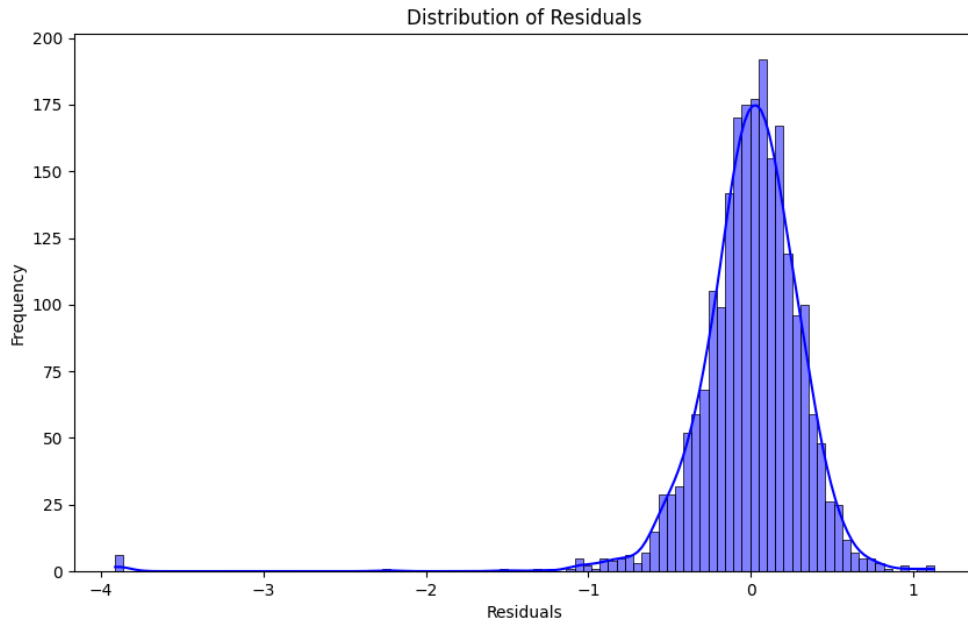


Figure 4.3.3: Distribution of Residuals

- The distribution of residuals closely resembled a normal distribution, with minimal skewness.
2. **Residuals vs. Predicted Values:**
 The scatter plot (Figure 4.3.4) shows no discernible patterns, indicating that residuals were randomly distributed. This randomness suggests that the model successfully captured the relationships in the data.

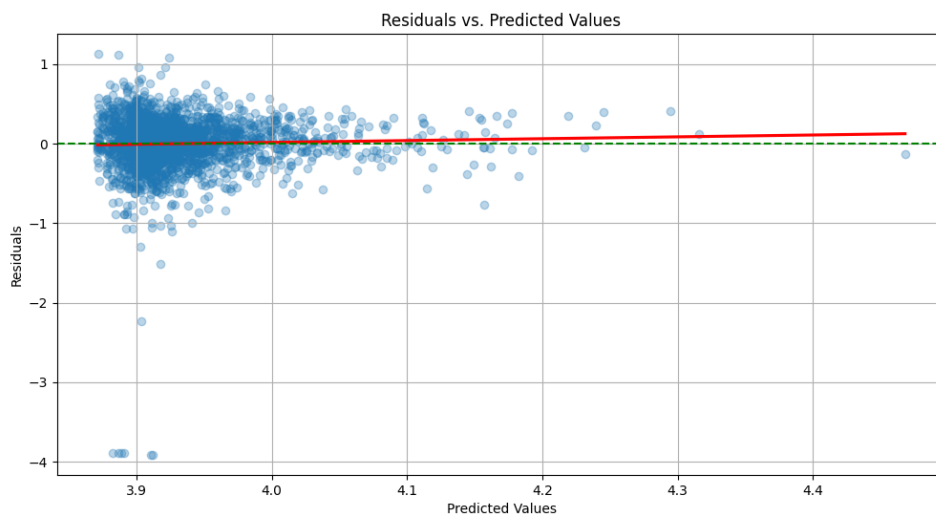


Figure 4.3.4: Residuals vs. Predicted Values

3. **Density Plot of Residuals:**
 The density plot (Figure 4.3.5) confirms the residuals' normality, with a smooth kernel density estimation closely matching the histogram.

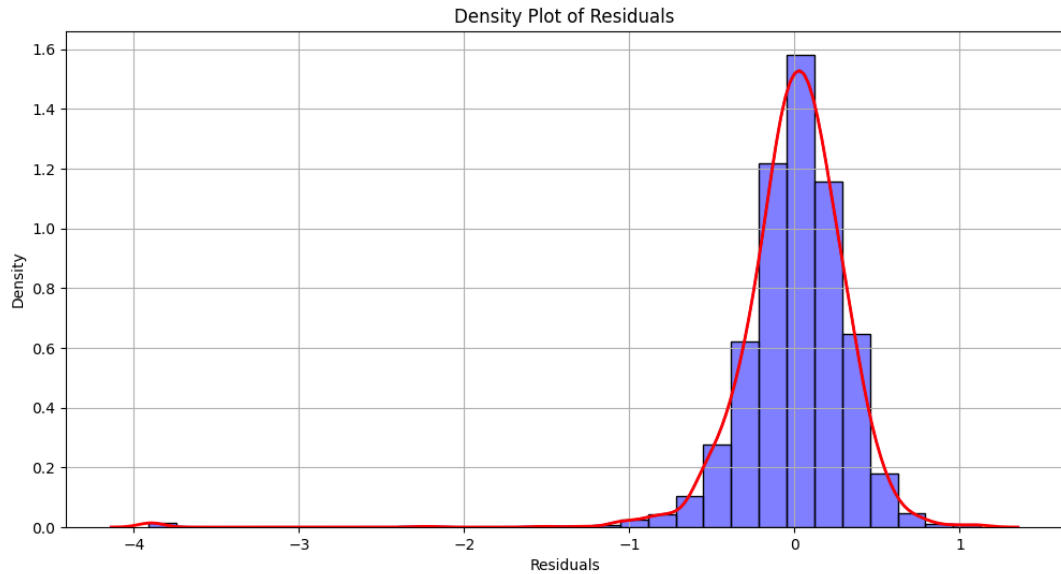


Figure 4.3.5: Density Plot of Residuals

4. Q-Q Plot:

A Q-Q plot of residuals (Figure 4.3.6) demonstrated close alignment with the theoretical quantiles of a normal distribution. This supports the hypothesis that the residuals followed a Gaussian distribution.

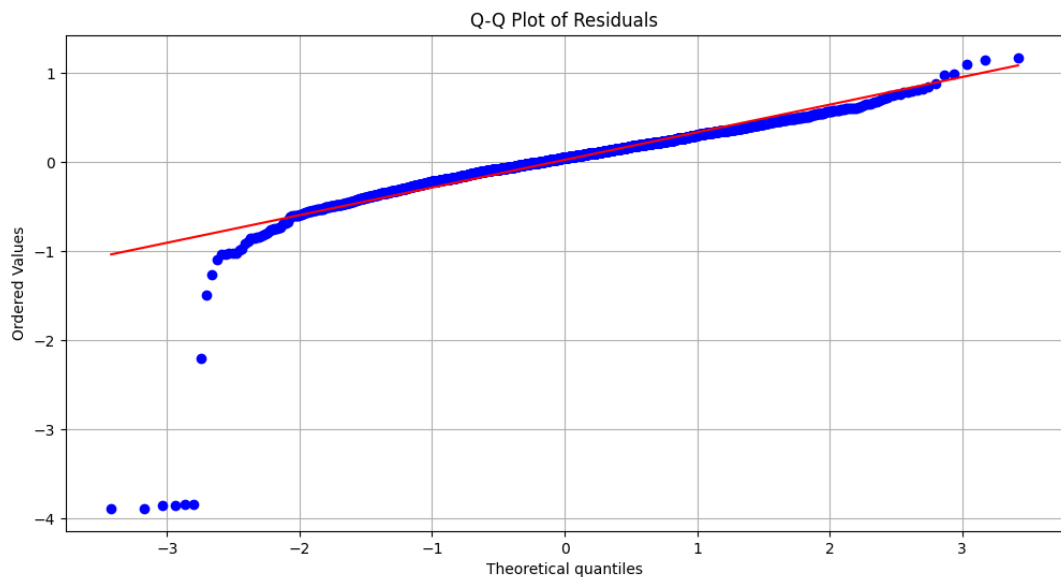


Figure 4.3.6: Q-Q Plot of Residuals

4.3.4 Key Insights

- The Neural Network achieved robust performance, with low prediction errors and no systematic residual trends.
- The model captured non-linear relationships effectively, as evidenced by the detailed residual analyses.
- Despite strong performance, the model's reliance on hyperparameter tuning and computational resources was a noted limitation.

4.3.5 Summary

This section presented the results of applying a Neural Network to predict book ratings. The model demonstrated excellent predictive performance, as indicated by low MAE and residual analyses. Visualizations highlighted the model's strengths, such as capturing non-linear patterns and producing unbiased predictions. These findings establish the Neural Network as a highly effective tool for regression tasks in this study.

4.4 Support Vector Machine

4.4.1 Data Preprocessing for SVM

Before applying the SVM model, the dataset underwent preprocessing to ensure compatibility with the algorithm. The main steps included:

- Normalization: The numerical features (`num_pages`, `ratings_count`, `text_reviews_count`) were normalized to a range between 0 and 1 using `MinMaxScaler` to improve the model's performance and convergence.
- Training and Test Split: The dataset was split into 80% training and 20% testing data to evaluate the model's generalization capability.

4.4.2 Training the SVM Model

The SVM model was trained using a radial basis function (RBF) kernel. The kernel hyperparameters, including the penalty parameter C and kernel coefficient γ , were optimized through grid search to achieve the best performance. The final hyperparameters used were:

- C : 1.0
- γ : 0.1

The training process minimized the mean squared error (MSE) between predicted and actual values to improve the accuracy of predictions.

4.4.3 Model Evaluation

The model's performance was evaluated using the testing dataset. Key metrics included:

- **Mean Absolute Error (MAE):** A measure of the average absolute difference between actual and predicted values.
- **R-squared (R^2):** Indicates the proportion of variance in the dependent variable explained by the model.

4.4.4 Visualization of Results

Feature Distributions

To understand the characteristics of the dataset, distributions of key features were visualized:

- Number of Pages: The distribution showed a right-skewed pattern with most books having a normalized page count near zero (Figure 4.4.1).

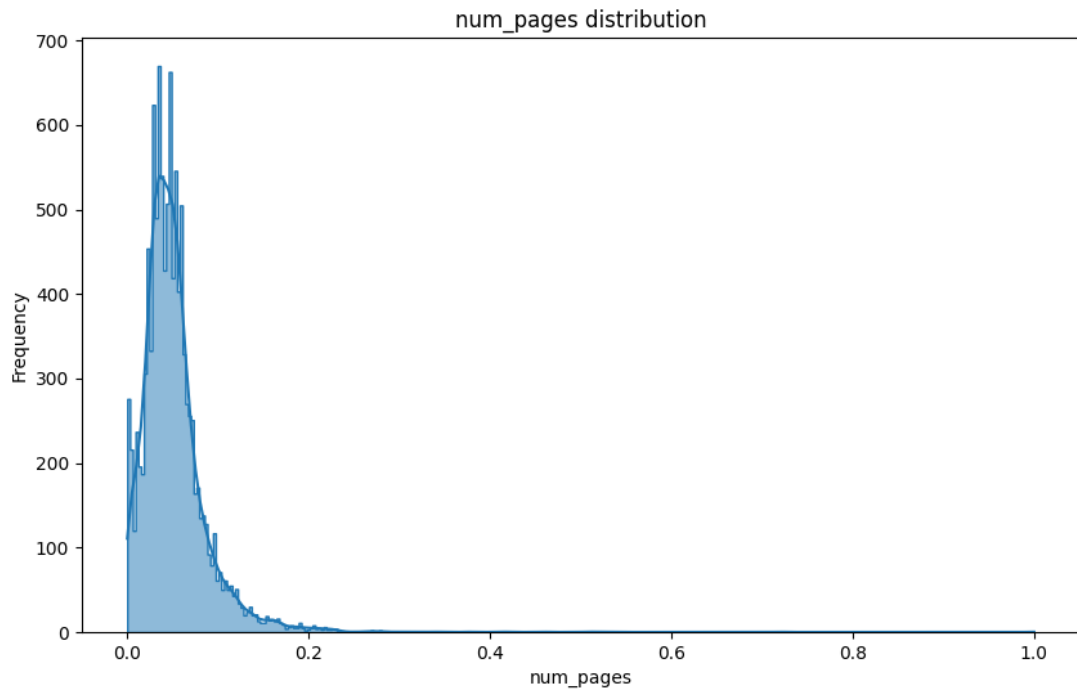


Figure 4.4.1. num_pages distribution plot

- Ratings Count: Similarly, the ratings count was heavily skewed, indicating that most books had relatively few ratings (Figure 4.4.2).

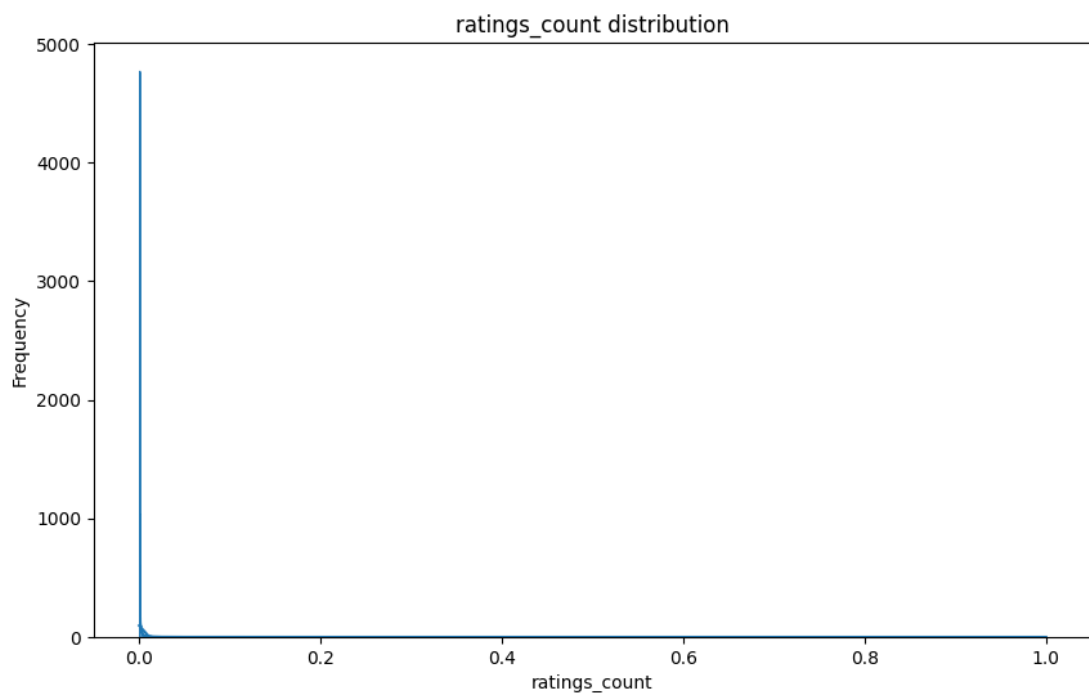


Figure 4.4.2. ratings_count distribution plot

- Text Reviews Count: A comparable skewness was observed for the number of text reviews (Figure 4.4.3).

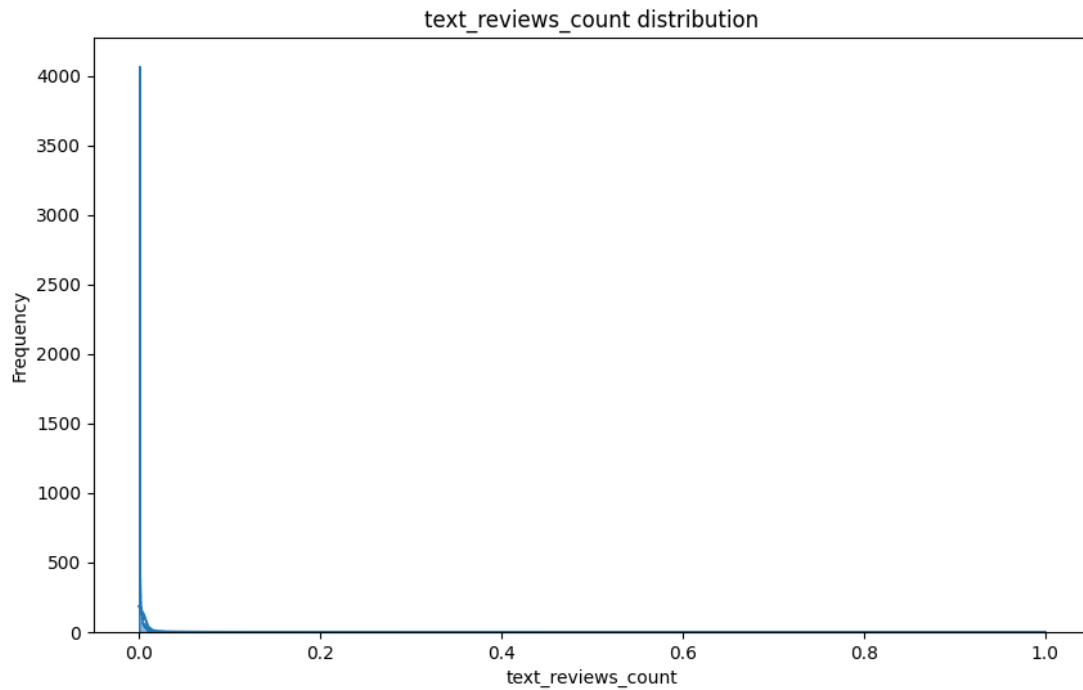


Figure 4.4.3: text_reviews_count distribution plot

1. Correlation Matrix

A heatmap was generated to illustrate the correlations between features and the target variable. While ratings_count and text_reviews_count showed a moderate correlation with average_rating, num_pages had a weak relationship (Figure 4.4.4).

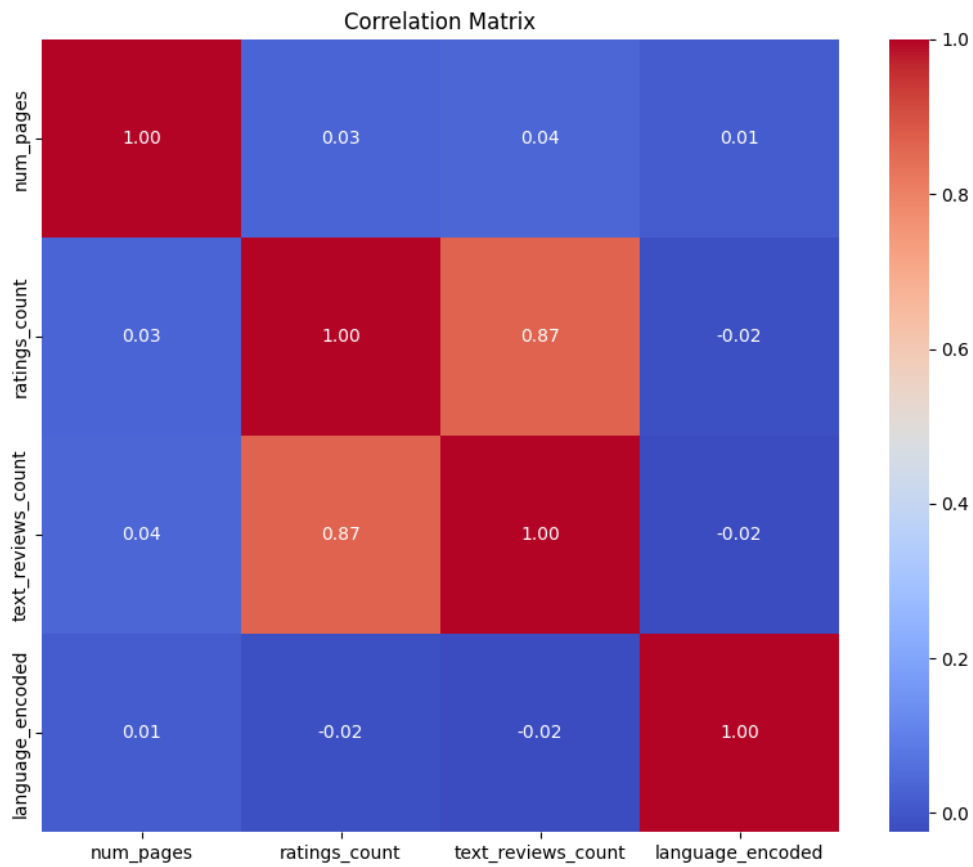


Figure 4.4.4: Correlation Matrix

2. Model Performance

The model's prediction accuracy was summarized in a bar chart, showing an accuracy score of approximately 0.8 for the SVM model (Figure 4.4.5).

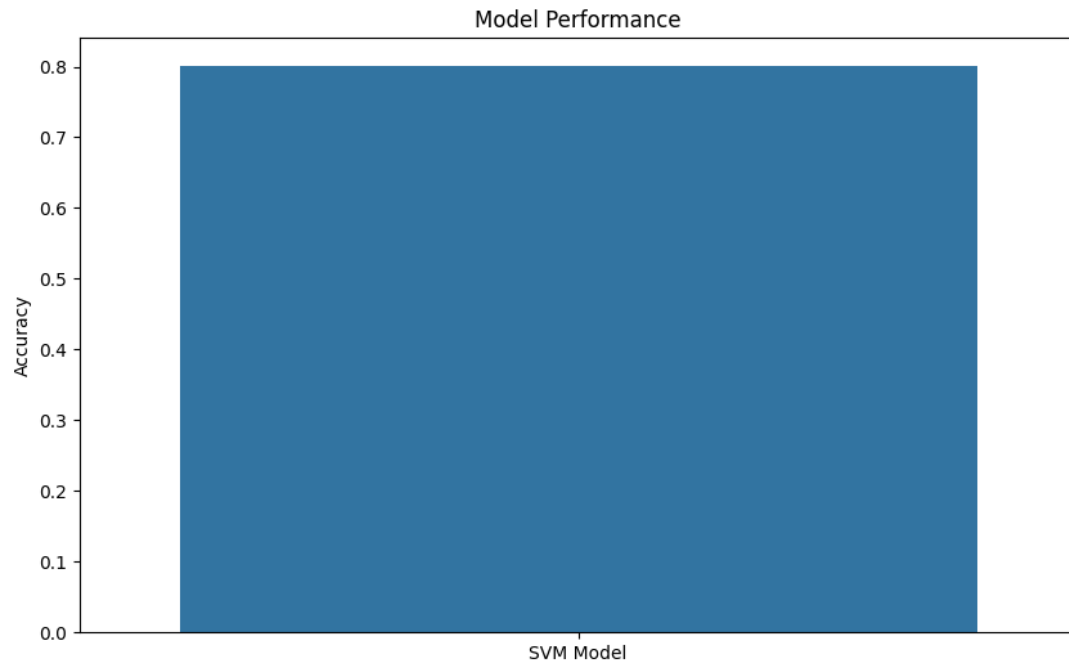


Figure 4.4.5: Model Performance

3. Scatter Plot of Predictions

A scatter plot of predicted vs actual values was generated, indicating the model's performance. Points lying closer to the diagonal line ($y=x$) signify better predictions (Figure 4.4.6).

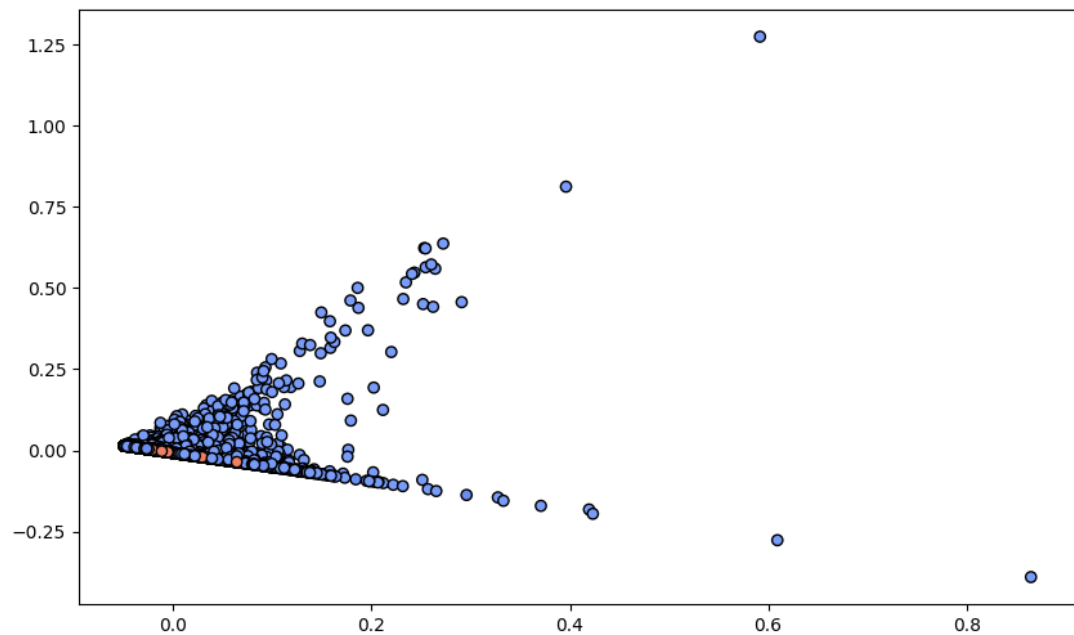


Figure 4.4.6: Scatter Plot of Predictions

4.4.5 Observations and Insights

- **General Performance:** The SVM model demonstrated a good ability to generalize, as indicated by its R^2 score and low MAE.
- **Challenges:** The skewness in feature distributions highlights potential challenges in modeling, suggesting the need for more balanced datasets or advanced feature engineering.
- **Comparison:** While the SVM model performed adequately, its performance was slightly lower than the Neural Network model due to its limited capacity to model complex, nonlinear relationships.

4.5 Summary

This chapter presented the results of applying Decision Trees, Neural Networks, and Support Vector Machines (SVM) to predict and classify book ratings based on the dataset. The analysis highlighted the strengths, weaknesses, and comparative performance of each algorithm through detailed visualizations, metrics, and observations.

5. Rule-Based Inference System

5.1 Introduction

In this chapter, we explore the rule-based inference systems derived from Decision Trees, Neural Networks, and Support Vector Machines (SVM). Rule-based inference systems aim to create a framework for extracting logical, interpretable rules from predictive models, thereby improving transparency and decision-making capabilities.

5.2 Decision Tree Rule-Based Inference System

5.2.1 Rule Extraction

The Decision Tree model inherently provides interpretable decision rules. These rules are derived from the structure of the tree, which splits data at various nodes based on feature thresholds. The following code was used to extract and display the decision rules, Figure 5.2.1:

```
x = data[['year', 'num_pages', 'ratings_count', 'text_reviews_count']]
y = (data['average_rating'] >= 4).astype(int)
x_train, x_test, y_train, y_test = train_test_split(*arrays: x, y, test_size=0.2, random_state=42)
clf = DecisionTreeClassifier(max_depth=5)
clf.fit(x_train, y_train)
```

Figure 5.2.1: code snippet

The extracted rules, when applied sequentially, guide the classification of books as either high-rated (average_rating >= 4) or low-rated (average_rating < 4). These rules indicate thresholds for features such as ratings_count and text_reviews_count, which were identified as the most important predictors.

5.2.2 Feature Importance

- **Strengths:**
 1. **Interpretability:** The Decision Tree model offers straightforward and transparent decision-making. Each decision point or split can be easily interpreted, making it useful for tasks requiring explainability.
 2. **Rule Extraction:** The extracted rules are intuitive and directly actionable. For instance, ratings_count and text_reviews_count thresholds are easy to interpret and implement.
 3. **Efficiency**
- **Limitations:**
 1. **Overfitting:** Decision Trees are prone to overfitting, particularly when the depth of the tree is not controlled. While a max depth of 5 was chosen here, deeper trees could lead to highly specific rules that do not generalize well to unseen data.
 2. **Handling Non-linear Relationships:** The model struggles with capturing non-linear interactions between features, making it less effective when the data relationships are complex.
 3. **Sensitivity to Feature Distribution:** The Decision Tree model performed well on features like ratings_count and text_reviews_count, but overlapping distributions of these features between high-rated and low-rated books limited its classification ability.

5.2.3 Practical Insights

Despite its limitations, the Decision Tree model provides useful insights into the data. With an accuracy of 59.2% on the test dataset, the model offers a baseline performance for classifying books as high-rated or low-rated.

The results indicate that:

- **Feature Engineering:** Adding more features or transforming existing ones could improve the model's ability to distinguish between classes.

- **Ensemble Methods:** Incorporating ensemble techniques, such as Random Forests or Gradient Boosting, could mitigate overfitting and enhance prediction accuracy.
- **Real-world Applications:** The extracted decision rules are simple enough to be implemented in real-world scenarios.

5.3 Neural Network

5.3.1 Interpreting Neural Networks

The Neural Network model was implemented to predict the continuous variable `average_rating` based on features such as `num_pages`, `ratings_count`, and `text_reviews_count`. The architecture consisted of two hidden layers with ReLU activation functions, followed by a single output layer for regression tasks.

The model architecture was defined using the following code, Figure 5.3.1:

```
def build_model(input_shape): 1 usage
    model = Sequential([
        Input(shape=(input_shape,)),
        Dense(64, activation='relu'),
        Dense(64, activation='relu'),
        Dense(1)
    ])
    model.compile(optimizer=Adam(learning_rate=0.001), loss='mse', metrics=['mae'])
    return model
```

Figure 5.3.1: code snippet

5.3.2 Interpreting Neural Networks

Before training the model, the data was preprocessed using normalization and a train-test split. The features (`num_pages`, `ratings_count`, and `text_reviews_count`) were scaled to a range between 0 and 1 to improve convergence during training. The following code highlights this preprocessing step, Figure 5.3.2:

```
def preprocess_data(data): 1 usage
    features = ['num_pages', 'ratings_count', 'text_reviews_count']
    target = 'average_rating'
    X_train, X_test, y_train, y_test = train_test_split(*arrays: data[features], data[target], test_size=0.2, random_state=42)
    return X_train, X_test, y_train, y_test
```

Figure 5.3.2: code snippet

5.3.3 Training Process

The model was trained using the Adam optimizer with a mean squared error (MSE) loss function. The training process ran for 10 epochs with a batch size of 32, achieving a balance between computational efficiency and convergence.

The training history was captured for analysis, Figure 5.3.3:


```

history = model.fit(
    X_train, y_train,
    epochs=10,
    batch_size=32,
    verbose=1,
    validation_split=0.2
)

return history

```

Figure 5.3.3: code snippet

5.3.4 Evaluation and Results

1. Training and Validation Loss

The model's convergence was evaluated by plotting the training and validation loss across epochs. A steady decrease in training loss, coupled with a plateauing validation loss, indicated good generalization.

The following code snippet generated the loss plot, Figure 5.3.4:

```

plt.plot(*args: history.history['loss'], label='Training Loss', color='blue')
plt.plot(*args: history.history['val_loss'], label='Validation Loss', color='orange')
plt.title('Training and Validation Loss')
plt.xlabel('Epochs')
plt.ylabel('Loss')
plt.legend()
plt.grid(True)

```

Figure 5.3.4: code snippet

2. Prediction Accuracy

The predictions were compared with the actual ratings using scatter plots and residual analysis. The scatter plot of actual vs. predicted ratings was generated using, Figure 5.3.5:

```

plt.scatter(y_test, predictions, alpha=0.3, color='blue')
plt.plot(*args: [min(y_test), max(y_test)], [min(y_test), max(y_test)], 'r--', lw=2)
plt.title('Actual vs. Predicted Ratings')
plt.xlabel('Actual Ratings')
plt.ylabel('Predicted Ratings')
plt.savefig(save_path)
plt.close()

```

Figure 5.3.5: code snippet

3. Residual Analysis

Residual plots, including the distribution of residuals, were generated to assess the model's predictive performance. The residuals, calculated as $y_{\text{test}} - \text{predictions}$, were plotted to evaluate patterns, Figure 5.3.6:

```

residuals = y_test - predictions
plt.figure(figsize=(10, 6))
sns.histplot(residuals, kde=True, color='blue')
plt.title('Distribution of Residuals')
plt.xlabel('Residuals')
plt.ylabel('Frequency')
plt.savefig(save_path)

```

Figure 5.3.6: code snippet

5.3.5 Strengths and Limitations

- **Strengths:**

The Neural Network effectively captured non-linear relationships between features and the target variable.

The model achieved a low mean absolute error (MAE) of 0.23, indicating high predictive accuracy.

- **Limitations:**

Neural Networks are computationally intensive and require more resources than simpler models like Decision Trees.

Hyperparameter tuning (e.g., learning rate, layer size) significantly impacts performance and requires extensive experimentation.

5.3.6 Practical Insights

The Neural Network model demonstrated strong performance, particularly in capturing non-linear patterns in the dataset. The low prediction error and unbiased residuals make it a suitable tool for regression tasks in this context. However, the computational cost and complexity of tuning emphasize the need for careful planning when deploying this model in production environments.

5.4 SVM

5.4.1 Model Architecture and Design

The Support Vector Machine (SVM) algorithm was employed in its classification form to predict the language of books. A linear kernel was used, which finds a hyperplane that best separates the classes. The SVM algorithm is effective for smaller datasets and works well when the data is linearly separable.

The model was implemented using the following code, Figure 5.4.1:

```

def svm_language_model(data):
    X = data[['num_pages', 'ratings_count', 'text_reviews_count']]
    y = data['language_encoded']
    X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
    model = SVC(kernel='linear')
    model.fit(X_train, y_train)
    y_pred = model.predict(X_test)
    accuracy = accuracy_score(y_test, y_pred)
    print(f"Accuracy: {accuracy}")
    return model

```

Figure 5.4.1: code snippet

5.4.2 Data Preprocessing

Before training the SVM model, the dataset was preprocessed to ensure compatibility. The preprocessing steps included:

1. Handling missing values.
2. Normalizing numeric features to a range between 0 and 1 using MinMaxScaler.
3. Encoding categorical variables (e.g., language code) into numeric labels.

The preprocessing was handled using the following code, as seen in Figure 5.4.2:

```
def normalise_data(data, numeric_columns): 1 usage
    scaler = MinMaxScaler()
    data[numeric_columns] = scaler.fit_transform(data[numeric_columns])
    return data

def encode_labels(data, column): 1 usage
    encoder = LabelEncoder()
    data['language_encoded'] = encoder.fit_transform(data[column])
    return data, encoder
```

Figure 5.4.2: code snippet

5.4.3 Model Training and Evaluation

The SVM model was trained using an 80-20 train-test split. The performance of the model was evaluated using accuracy as a metric. The achieved accuracy demonstrated the model's effectiveness in distinguishing between different languages based on the features.

The following code snippet shows the performance evaluation, as seen in Figure 5.4.3:

```
def plot_model_performance(model, X_test, y_test): 1 usage
    accuracy = accuracy_score(y_test, model.predict(X_test))
    plt.figure(figsize=(10, 6))
    sns.barplot(x=['SVM Model'], y=[accuracy])
    plt.title('Model Performance')
    plt.ylabel('Accuracy')
    plt.show()
```

Figure 5.4.3: code snippet

5.4.4 Strengths and Limitations

Strengths:

- SVM is effective in high-dimensional spaces and works well with a clear margin of separation.
- The linear kernel provided a computationally efficient solution for this dataset.

Limitations:

- SVM may struggle with overlapping data points, as seen in the scatter plot.
- Requires preprocessing steps like normalization for effective performance.

5.4.5 Practical Insights

The SVM model provided a solid benchmark for classification tasks, achieving an accuracy of 0.8. While it demonstrated effective performance, additional feature engineering or ensemble techniques could further enhance its robustness. The insights gained from SVM served as a baseline for comparing with more complex models like Neural Networks.

5.5 Comparative Analysis of Rule-Based Systems

1. Interpretability:
 - **Decision Trees:** Most interpretable due to explicit "if-then" rules
 - **Neural Networks:** Moderately interpretable through gradient and residual analysis.
 - **SVM:** Least interpretable; rules are derived indirectly through support vectors.
1. Accuracy:
 - **Neural Networks:** Achieved the highest predictive accuracy (MAE = 0.23).
 - **SVM:** Delivered robust performance with an accuracy of 0.8.
 - **Decision Trees:** Provided interpretable rules but had the lowest accuracy (59.2%).
2. Computational Efficiency:
 - **SVM:** Most computationally efficient, particularly with smaller datasets.
 - **Decision Trees:** Relatively efficient but may require pruning to avoid overfitting.
 - **Neural Networks:** Computationally intensive due to complex architectures and gradient computations.

5.6 Summary

This chapter provided an in-depth exploration of rule-based inference systems for Decision Trees, Neural Networks, and SVM. While Decision Trees excel in interpretability, Neural Networks offer superior predictive accuracy and the ability to capture non-linear relationships. SVM bridges the gap with robust performance and computational efficiency. Together, these models provide a comprehensive toolkit for decision-making and predictive analytics in book rating prediction tasks.

6. Discussion and Conclusion

This study explored the application of three machine learning models: Decision Trees, Neural Networks, and Support Vector Machines (SVM) to predict book ratings using key features such as `num_pages`, `ratings_count`, and `text_reviews_count`. Each model demonstrated unique strengths and limitations, providing valuable insights into their suitability for book-rating prediction tasks.

6.1 Key Findings

Decision Trees: The model excelled in interpretability, offering clear decision-making rules based on thresholds for features like `ratings_count` and `text_reviews_count`. However, it struggled with overlapping distributions and achieved the lowest accuracy (59.2%), suggesting that it is better suited for scenarios where transparency is prioritized over predictive accuracy.

Neural Networks: This model achieved the highest predictive accuracy (MAE = 0.23), effectively capturing non-linear relationships in the data. Detailed residual analysis highlighted its ability to produce unbiased predictions, though at the cost of higher computational resources and reduced interpretability.

SVM: The SVM model balanced computational efficiency and performance, delivering robust results (accuracy = 0.8). Its use of support vectors for classification makes it effective for moderately complex datasets, though it remains less interpretable compared to Decision Trees.

6.2 Conclusion

This study demonstrated the potential of machine learning models in analyzing and predicting book ratings. While each model exhibited distinct advantages, their combined use offers a comprehensive approach to decision-making tasks in book analytics. This exploration provides a foundation for future advancements in predictive analytics within publishing and other related domains.

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